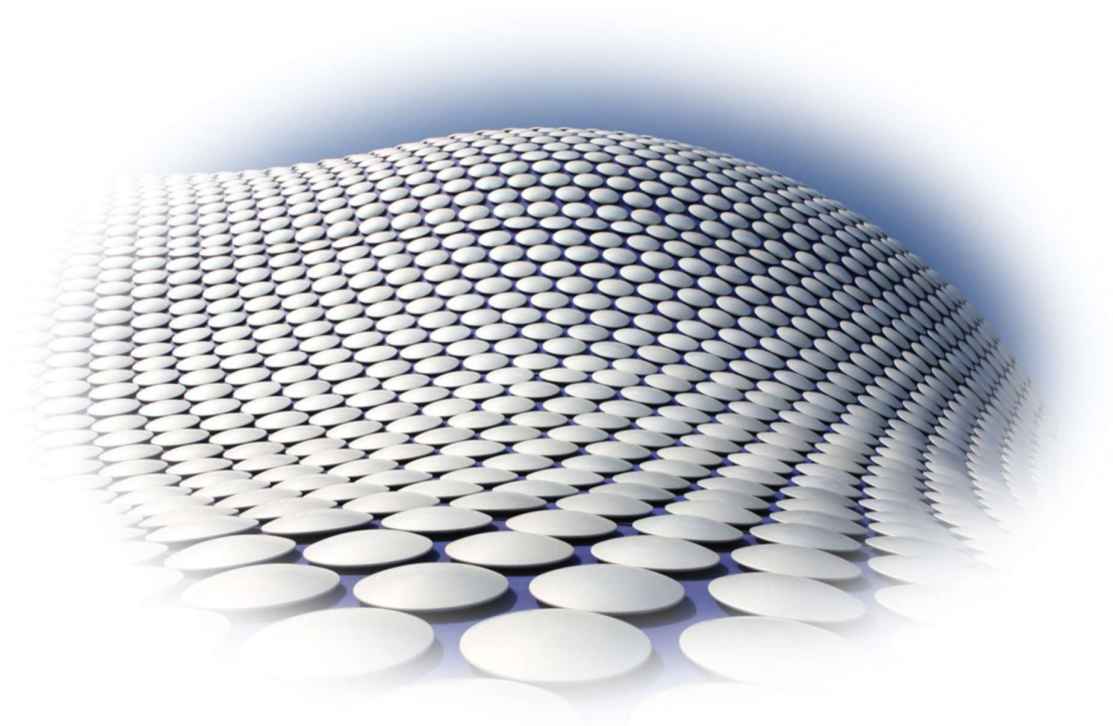




University Preparatory Program (Mathematics) Partial Differentiation



Engineering Mathematics Team

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5. Partial Differentiation

Learning Objectives

Upon completion of this lesson, you should be able to:

1. Calculate partial derivatives of a multi-variable function
2. Apply chain rule to find partial derivatives
3. Calculate directional derivatives and the gradient
4. Construct the equation of normal lines and tangent planes
5. Compute the relative extrema of functions of two-variables

5.1 Multi-variable Functions and Partial Derivatives

A **multi-variable function** is a function that produces one output from two or more inputs. The input variables are called independent variables and the output variable is called the dependent variable.

For instance, $z = \sqrt{1 - x^2 - y^2}$ is a two-variable function with independent (input) variables x and y , and dependent (output) variable z . In general, we write a two-variable function as $z = f(x, y)$.

For our purpose, we consider two-variable functions of real values, i.e. functions of the form $f: \mathbb{R}^2 \rightarrow \mathbb{R}$. The domain of a two-variable function f is the collection of all possible ordered pairs (x, y) that produce real-valued outputs $f(x, y)$. For instance, the domain of the two-variable function $z = \sqrt{1 - x^2 - y^2}$ is the unit disc $\{(x, y) \in \mathbb{R}^2: x^2 + y^2 \leq 1\}$.

The graph of a two-variable function $z = f(x, y)$ is a surface in 3D space, as shown in Figure 5.1.1.

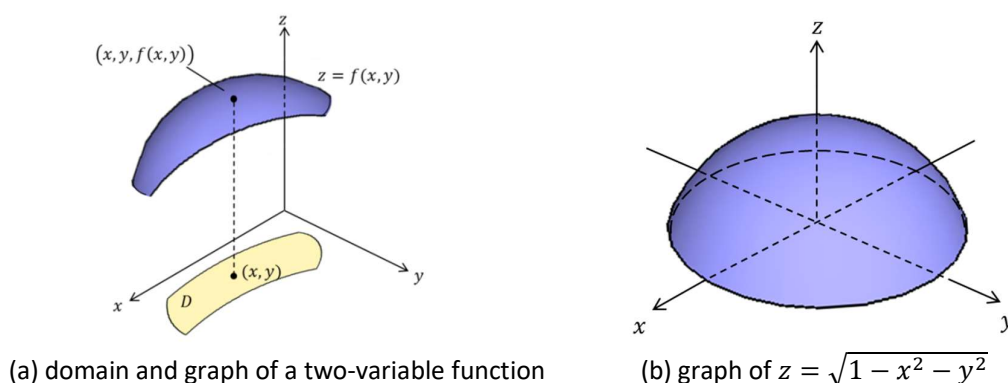


Figure 5.1.1. Graph of two-variable functions as surfaces in $\mathbb{R}^3$

A **partial derivative** of a multi-variable function f is the derivative of f with respect to one of the input variable, while keeping the other input variables fixed. For instance, a two-variable function $f(x, y)$ has two partial derivatives:

1. partial derivative of f with respect to x (keep the y -variable fixed), denoted by $\frac{\partial f}{\partial x}$ or f_x
2. partial derivative of f with respect to y (keep the x -variable fixed), denoted by $\frac{\partial f}{\partial y}$ or f_y

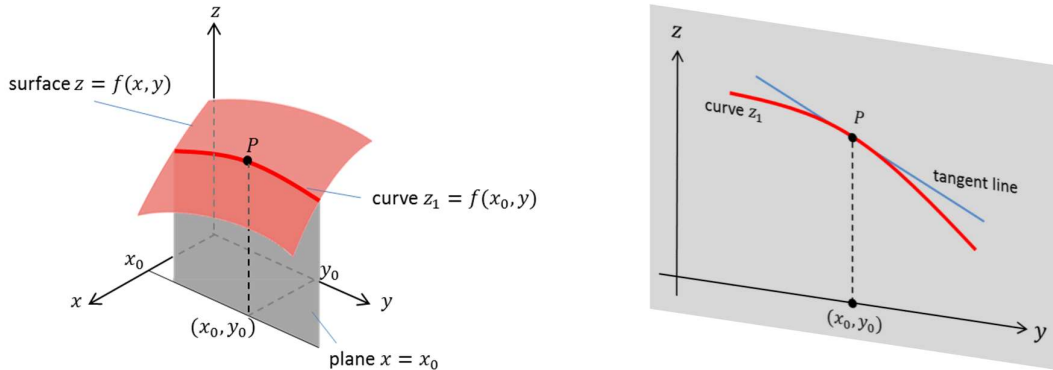
Example 5.1.1

Given $f(x, y) = \sqrt{1 - x^2 - y^2}$, calculate the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \left(\left(\underbrace{1 - y^2}_{\text{fixed}} - x^2 \right)^{1/2} \right) = \frac{1}{2} (1 - y^2 - x^2)^{-1/2} \cdot \frac{\partial}{\partial x} (1 - y^2 - x^2) = \frac{-x}{\sqrt{1 - y^2 - x^2}}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \left(\left(\underbrace{1 - x^2}_{\text{fixed}} - y^2 \right)^{1/2} \right) = \frac{1}{2} (1 - x^2 - y^2)^{-1/2} \cdot \frac{\partial}{\partial y} (1 - x^2 - y^2) = \frac{-y}{\sqrt{1 - x^2 - y^2}}$$

Consider a two-variable function $z = f(x, y)$. We shall discuss the geometric meaning of the partial derivative $\frac{\partial f}{\partial y}$ at the point $P(x_0, y_0, f(x_0, y_0))$. In this case, we keep the x -variable fixed, i.e. $x = x_0$. Figure 5.1.2(a) shows the surface of $z = f(x, y)$, the plane $x = x_0$, and the intersection curve, $z_1 = f(x_0, y)$, which can be regarded as the graph of a single-variable function of input variable y and output variable z . Hence, the partial derivative $\frac{\partial f}{\partial y}$ at (x_0, y_0) is the slope of the tangent line of the curve z_1 at $y = y_0$, as shown in Figure 5.1.2(b).



(a) intersection of $z = f(x, y)$ and $x = x_0$

(b) curve $z_1 = f(x_0, y)$ on the plane $x = x_0$

Figure 5.1.2. Geometric meaning of a partial derivative of a two-variable function

Similar observation can be made for the partial derivative $\frac{\partial f}{\partial x}$. The following notations are used to denote the values of the partial derivatives at the point $(x_0, y_0, f(x_0, y_0))$:

1. partial derivative with respect to x : $\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}$ or $f_x(x_0, y_0)$
2. partial derivative with respect to y : $\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)}$ or $f_y(x_0, y_0)$

Linear properties, as well as product and quotient rules, for partial derivatives are similar to those for ordinary derivatives. However, the chain rule for partial derivatives is a little more complex.

Consider a function $z = f(x, y)$ for which its partial derivatives exist. If $x = g(u, v)$ and $y = h(u, v)$ are functions of u and v , for which their partial derivatives exist, then the partial derivatives of z with respect to u and v are given as follows.

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \quad (5.1)$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v} \quad (5.2)$$

The formulas (5.1) and (5.2) are known as the chain rule for two independent parameters u and v . These can be easily remembered from the diagram shown in Figure 5.1.3. For instance, (5.1) is obtained by tracing two branches from the variable z down to the variable u . The chain rule can be extended to n numbers of independent parameters.

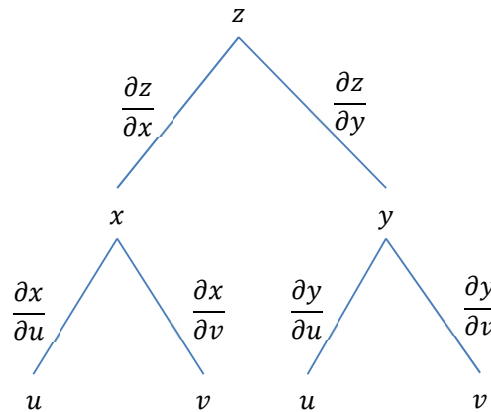


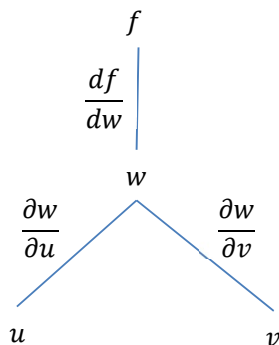
Figure 5.1.3. Chain rule diagram for two independent parameters u and v

Example 5.1.2

If $z = u + f(uv)$, show that $u \frac{\partial z}{\partial u} - v \frac{\partial z}{\partial v} = u$.

$z = u + f(uv)$ implies $\frac{\partial z}{\partial u} = 1 + \frac{\partial f}{\partial u}$ and $\frac{\partial z}{\partial v} = \frac{\partial f}{\partial v}$. Let $w = uv$. Then $\frac{\partial w}{\partial u} = v$ and $\frac{\partial w}{\partial v} = u$.

We apply chain rule to find f_u and f_v .



$$\text{So, } \frac{\partial z}{\partial u} = 1 + \frac{df}{dw} \frac{\partial w}{\partial u} = 1 + f'(w)v$$

$$\text{and } \frac{\partial z}{\partial v} = \frac{df}{dw} \frac{\partial w}{\partial v} = f'(w)u$$

$$\text{Thus, } u \frac{\partial z}{\partial u} - v \frac{\partial z}{\partial v} = u(1 + f'(w)v) - v f'(w)u = u$$

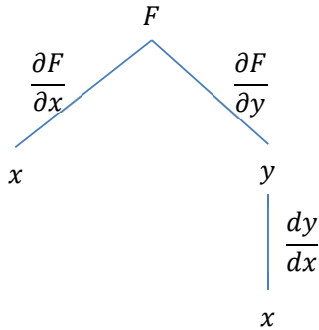
The chain rule for partial derivatives can be used to find the derivative $\frac{dy}{dx}$, where y is implicitly defined as a function of x .

Theorem 5.1.3

If $F(x, y) = 0$ defines y implicitly as a differentiable function of x , then

$$\frac{dy}{dx} = \frac{-F_x}{F_y} \text{ provided } F_y \neq 0$$

Proof.



$$\begin{aligned} F(x, y) = 0 &\Rightarrow \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0 \\ &\Rightarrow F_y \frac{dy}{dx} = -F_x \\ &\Rightarrow \frac{dy}{dx} = \frac{-F_x}{F_y} \end{aligned}$$

Example 5.1.4

The lemniscate of Bernoulli is given by the equation $(x^2 + y^2)^2 = 9(x^2 - y^2)$, find $\frac{dy}{dx}\big|_{(\sqrt{2}, 1)}$.

Let $F(x, y) = (x^2 + y^2)^2 - 9(x^2 - y^2)$. Then

$$\begin{aligned} F_x &= \frac{\partial((x^2 + y^2)^2)}{\partial x} - 9 \frac{\partial(x^2 - y^2)}{\partial x} \\ &= 2(x^2 + y^2) \frac{\partial(x^2 + y^2)}{\partial x} - 9(2x) \\ &= 2x(2x^2 + 2y^2 - 9) \end{aligned}$$

$$\begin{aligned} F_y &= \frac{\partial((x^2 + y^2)^2)}{\partial y} - 9 \frac{\partial(x^2 - y^2)}{\partial y} \\ &= 2(x^2 + y^2) \frac{\partial(x^2 + y^2)}{\partial y} - 9(-2y) \\ &= 2y(2x^2 + 2y^2 + 9) \end{aligned}$$

$$\frac{dy}{dx}\bigg|_{(\sqrt{2}, 1)} = \frac{-F_x}{F_y}\bigg|_{(\sqrt{2}, 1)} = \frac{-2x(2x^2 + 2y^2 - 9)}{2y(2x^2 + 2y^2 + 9)}\bigg|_{x=\sqrt{2}, y=1} = \frac{-\sqrt{2}(4 + 2 - 9)}{4 + 2 + 9} = \frac{\sqrt{2}}{5}$$

The second order partial derivatives are defined similarly to that of ordinary derivatives.

$$f_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} \quad \text{and} \quad f_{yy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2}$$

Moreover, we have the following mixed second order partial derivatives.

$$f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} \quad \text{and} \quad f_{yx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y}$$

In general, $f_{xy} \neq f_{yx}$. However, the equality of mixed second order partial derivatives can be obtained under certain circumstances.

Theorem 5.1.5

If the function $f(x, y)$ has mixed second order partial derivatives f_{xy} and f_{yx} that are *continuous* in an open disc containing (x_0, y_0) , then $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$.

Example 5.1.6

Given $f(x, y) = xy^2 + 2xy + x^2y$, calculate f_{xx} , f_{yy} , f_{xy} , and f_{yx} .

$$f_x = \frac{\partial}{\partial x} (xy^2 + 2xy + x^2y) = y^2 + 2y + 2xy$$

$$f_y = \frac{\partial}{\partial y} (xy^2 + 2xy + x^2y) = 2xy + 2x + x^2$$

$$f_{xx} = \frac{\partial}{\partial x} (f_x) = \frac{\partial}{\partial x} (y^2 + 2y + 2xy) = 2y$$

$$f_{yy} = \frac{\partial}{\partial y} (f_y) = \frac{\partial}{\partial y} (2xy + 2x + x^2) = 2x$$

$$f_{xy} = \frac{\partial}{\partial y} (f_x) = \frac{\partial}{\partial y} (y^2 + 2y + 2xy) = 2y + 2 + 2x$$

$$f_{yx} = \frac{\partial}{\partial x} (f_y) = \frac{\partial}{\partial x} (2xy + 2x + x^2) = 2y + 2 + 2x$$

5.2 Directional Derivatives and the Gradient

Given a two-variable function $f(x, y)$, its partial derivatives can be written as dot products of vectors.

$$f_x = \begin{bmatrix} f_x \\ f_y \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} f_x \\ f_y \end{bmatrix} \cdot \hat{i} \quad \text{and} \quad f_y = \begin{bmatrix} f_x \\ f_y \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} f_x \\ f_y \end{bmatrix} \cdot \hat{j}$$

Geometrically, f_x is the rate of change of f in the direction of the positive x -axis and f_y is the rate of change of f in the direction of the positive y -axis. This can be generalized to determine the rate of change of f in the direction of a unit vector $\hat{\mathbf{u}} = u_1 \hat{\mathbf{i}} + u_2 \hat{\mathbf{j}}$, which is defined as the **directional derivative** of f in the direction of $\hat{\mathbf{u}}$, given as follows.

Directional Derivative

$$D_{\mathbf{u}}f = f_x u_1 + f_y u_2 = \begin{bmatrix} f_x \\ f_y \end{bmatrix} \cdot \hat{\mathbf{u}} \quad (5.3)$$

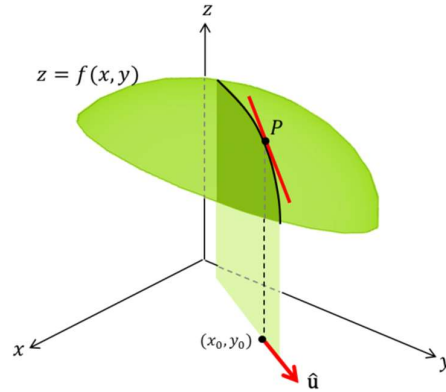


Figure 5.2.1. Directional derivative as the rate of change in the direction of a unit vector $\hat{\mathbf{u}}$

Example 5.2.1

Calculate the directional derivative of $f(x, y) = 5 - 3x^2 + y^3$ at $(2, 1)$ in the direction of $\begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$.

Let $\mathbf{u} = \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$. Then $\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|} = \frac{1}{2} \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$. Partial derivatives of f are $f_x = -6x$ and $f_y = 3y^2$.

$$D_{\mathbf{u}}f(2, 1) = \begin{bmatrix} f_x(2, 1) \\ f_y(2, 1) \end{bmatrix} \cdot \frac{1}{2} \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -6(2) \\ 3(1^2) \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix} = \frac{1}{2}(-12 - 3\sqrt{3}) \approx -8.598$$

We have seen the importance of the vector $\begin{bmatrix} f_x \\ f_y \end{bmatrix}$ in the calculation directional derivative. This vector is called the **gradient vector** of f , constructed by putting together all partial derivatives of $f(x, y)$.

Definition 5.2.2 (Gradient and Del Operator)

Given a two-variable function $f(x, y)$ for which its partial derivatives exist. The **gradient** of f , denoted by ∇f (read: “del eff”), is a vector given as

$$\nabla f = \begin{bmatrix} f_x \\ f_y \end{bmatrix}$$

∇ is called **del operator**, a generalization of the derivative operator for multi-variable functions.

We can now write $D_{\mathbf{u}}f = \nabla f \cdot \hat{\mathbf{u}}$. The gradient and del operator can be extended to multi-variable functions with n inputs. For instance, given $F(x, y, z)$, the gradient of F is

$$\nabla F = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}$$

Example 5.2.3

Given $F(x, y, z) = x \cos yz$, calculate ∇F and the directional derivative of F at $(1, 2, 0)$ in the direction of $\mathbf{v} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$.

$$\nabla F = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x}(x \cos yz) \\ \frac{\partial}{\partial y}(x \cos yz) \\ \frac{\partial}{\partial z}(x \cos yz) \end{bmatrix} = \begin{bmatrix} \cos yz \\ -xz \sin yz \\ -xy \sin yz \end{bmatrix}$$

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \Rightarrow \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

$$D_{\mathbf{v}}F(1, 2, 0) = \begin{bmatrix} \cos 0 \\ 0 \\ -2 \sin 0 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \frac{1}{3}.$$

5.3 Normal Lines and Tangent Planes

Given the point $P(x_0, y_0, f(x_0, y_0))$ on the surface $z = f(x, y)$, we consider the plane $x = x_0$ and the curve of intersection $z_1 = f(x_0, y)$ between the surface and the plane, cf. Figure 5.1.1(a). The equation of the tangent line T_y of this curve at P is given by

$$z - f(x_0, y_0) = f_y(x_0, y_0)(y - y_0)$$

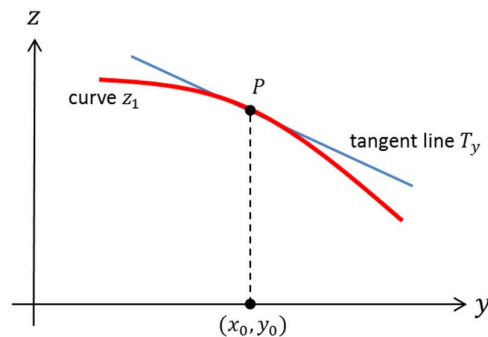


Figure 5.3.1. Tangent line T_y of the curve $z_1 = f(x_0, y)$ at P on the plane $x = x_0$

Using the notion for lines in 3D space, the symmetric equation of the tangent line T_y is

$$\frac{z - f(x_0, y_0)}{f_y(x_0, y_0)} = \frac{y - y_0}{1}, \quad x = x_0$$

In other words, the direction vector for the tangent line T_y is $\begin{bmatrix} 0 \\ 1 \\ f_y(x_0, y_0) \end{bmatrix}$.

Similarly, if we consider the intersection curve $z_2 = f(x, y_0)$ of the surface $z = f(x, y)$ and the plane $y = y_0$, the tangent line T_x of this curve at P has the symmetric equation

$$\frac{z - f(x_0, y_0)}{f_x(x_0, y_0)} = \frac{x - x_0}{1}, \quad y = y_0$$

In other words, the direction vector for the tangent line T_x is $\begin{bmatrix} 1 \\ 0 \\ f_x(x_0, y_0) \end{bmatrix}$.

The **tangent plane** to the surface $z = f(x, y)$ at P is the plane that contains both the tangent lines T_y and T_x . Hence, the normal vector $\mathbf{N}$ for the tangent plane is given as follows.

$$\mathbf{N} = \begin{bmatrix} 0 \\ 1 \\ f_y(x_0, y_0) \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ f_x(x_0, y_0) \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 1 & f_y(x_0, y_0) \\ 1 & 0 & f_x(x_0, y_0) \end{vmatrix} = \begin{bmatrix} f_x(x_0, y_0) \\ f_y(x_0, y_0) \\ -1 \end{bmatrix} \quad (5.4)$$

And thus, the equation of tangent plane of the surface $z = f(x, y)$ at P is given as

$$\begin{bmatrix} f_x(x_0, y_0) \\ f_y(x_0, y_0) \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - f(x_0, y_0) \end{bmatrix} = 0$$

The point-normal form of the tangent plane is given as follows.

$$\textbf{Tangent Plane} \quad z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \quad (5.5)$$

The **normal line** to the surface $z = f(x, y)$ at the point $P(x_0, y_0, f(x_0, y_0))$ is the line that is orthogonal to the tangent plane at P , i.e. the line with direction vector $\begin{bmatrix} f_x(x_0, y_0) \\ f_y(x_0, y_0) \\ -1 \end{bmatrix}$. The equation of this normal line is given as follows.

Normal Line

$$\begin{cases} x = x_0 + f_x(x_0, y_0)t \\ y = y_0 + f_y(x_0, y_0)t \\ z = f(x_0, y_0) - t \end{cases} \quad \text{for } t \in \mathbb{R} \quad (5.6)$$

$$\text{or } \frac{x - x_0}{f_x(x_0, y_0)} = \frac{y - y_0}{f_y(x_0, y_0)} = \frac{z - z_0}{-1} \quad (5.7)$$

Example 5.3.1

Construct the standard equation of the tangent plane and symmetric equation for the normal line to the surface $z = 9 - x^2 - y^2$ at the point $(1, 2, 4)$.

$$z_x = \frac{\partial}{\partial x}(9 - x^2 - y^2) = -2x$$

$$z_y = \frac{\partial}{\partial y}(9 - x^2 - y^2) = -2y$$

The normal vector at $(1, 2, 4)$ is $\mathbf{N} = \begin{bmatrix} z_x(1, 2) \\ z_y(1, 2) \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ -1 \end{bmatrix}$.

Standard equation of the tangent plane at $(1, 2, 4)$ is $\begin{bmatrix} -2 \\ -4 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x - 1 \\ y - 2 \\ z - 4 \end{bmatrix} = 0 \Rightarrow 2x + 4y + z - 14 = 0$.

Symmetric equation of the normal line at $(1, 2, 4)$ is $\frac{x - 1}{-2} = \frac{y - 2}{-4} = \frac{z - 4}{-1}$.

We would like to extend the construction of tangent planes and normal lines for *implicitly*-defined 3D surfaces, represented by the equation $F(x, y, z) = 0$. For example, the sphere $x^2 + y^2 + z^2 = 4$ is an implicitly-defined 3D surface that is represented by $x^2 + y^2 + z^2 - 4 = 0$. In contrast, the surface defined by $z = \sqrt{4 - x^2 - y^2}$ only gives the upper hemisphere, as shown in Figure 5.3.2.

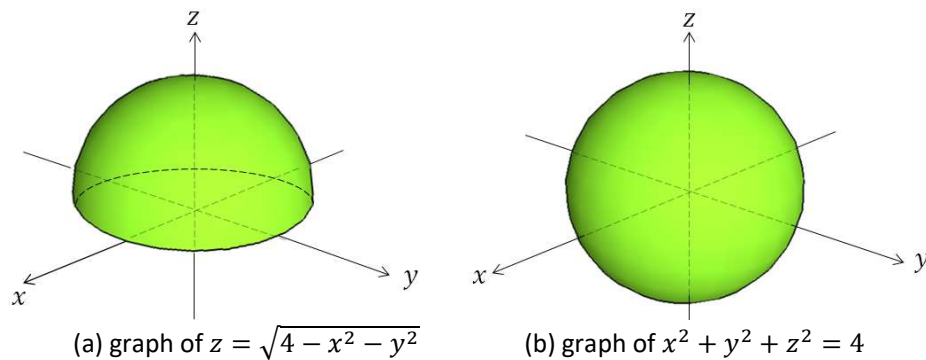


Figure 5.3.2. Explicitly- vs. implicitly-defined 3D surfaces

We observe that 3D surfaces that are explicitly defined by $z = f(x, y)$ can be represented as $F(x, y, z) = 0$, where $F(x, y, z) = f(x, y) - z$. In this case, the gradient of F is given as

$$\nabla F = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} f_x \\ f_y \\ -1 \end{bmatrix},$$

which implies that $\nabla F(x_0, y_0, z_0)$ is the normal vector of the surface of $F(x, y, z) = 0$ at the point (x_0, y_0, z_0) . This observation is generalized in the following theorem.

Theorem 5.3.2 (The Normal Property of the Gradient)

Let $P(x_0, y_0, z_0)$ be a point on the 3D surface $F(x, y, z) = 0$. Suppose the partial derivatives F_x , F_y , and F_z exist at P . If $\nabla F(x_0, y_0, z_0) \neq 0$, then it is orthogonal to the surface at P .

Example 5.3.3

Construct the standard equation of the tangent plane and symmetric equation for the normal line to the surface $x^2 + y^2 + z^2 = 4$ at the point $(1, 1, \sqrt{2})$.

Let $F(x, y, z) = x^2 + y^2 + z^2 - 4$. Let $\mathbf{N}$ be the normal vector to the sphere at $(1, 1, \sqrt{2})$.

$$\nabla F = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \\ 2z \end{bmatrix}$$

$$\Rightarrow \mathbf{N} = \nabla F(1, 1, \sqrt{2}) = \begin{bmatrix} 2 \\ 2 \\ 2\sqrt{2} \end{bmatrix}$$

Standard equation of the tangent plane at $(1, 1, \sqrt{2})$ is $2(x - 1) + 2(y - 1) + 2\sqrt{2}(z - \sqrt{2}) = 0$

$$\Rightarrow x + y + \sqrt{2}z - 4 = 0.$$

Symmetric equation of the normal line at $(1, 1, \sqrt{2})$ is $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-\sqrt{2}}{2\sqrt{2}}$.

5.4 Extrema of Two-Variable Functions

For a differentiable single-variable function $y = f(x)$, the zero derivative condition $f' = 0$, and the sign of its second derivative f'' enable us to determine the relative extrema of the function. This method will be extended to two-variable functions $z = f(x, y)$.

Definition 5.4.1 (Relative Extrema and Saddle Point of a Two-Variable Function)

Given a two-variable function $f(x, y)$ and a point (x_0, y_0) in its domain, f is said to have:

1. a **relative minimum** at (x_0, y_0) if $f(x_0, y_0) \leq f(x, y)$ for all (x, y) in an open disc containing (x_0, y_0) ;
2. a **relative maximum** at (x_0, y_0) if $f(x_0, y_0) \geq f(x, y)$ for all (x, y) in an open disc containing (x_0, y_0) ;
3. a **saddle point** at (x_0, y_0) if every open disc centered at (x_0, y_0) contains points (x_1, y_1) that satisfy $f(x_0, y_0) < f(x_1, y_1)$ as well as points (x_2, y_2) that satisfy $f(x_0, y_0) > f(x_2, y_2)$.

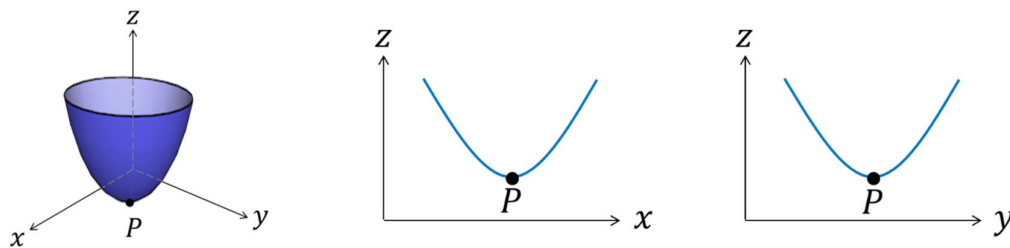


Figure 5.4.1. Relative minimum P of a two-variable function

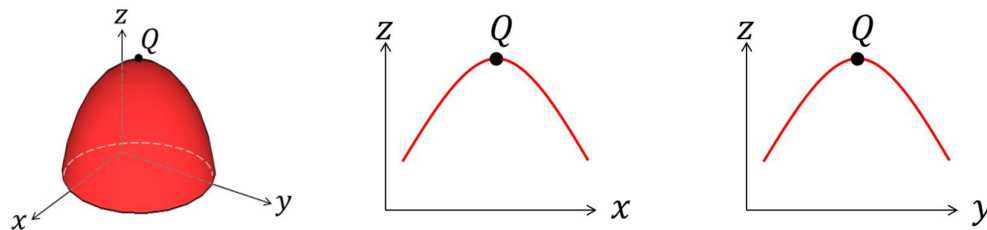


Figure 5.4.2. Relative maximum Q of a two-variable function

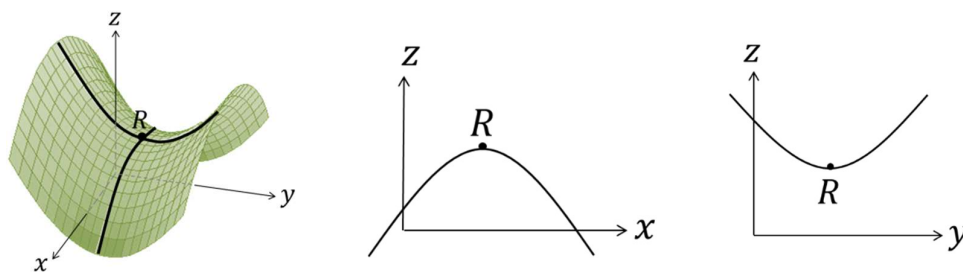


Figure 5.4.3. Saddle point R of a two-variable function

The zero derivatives condition for two-variable functions is as follows.

Theorem 5.4.2 (Zero Derivatives Condition)

Let $f(x, y)$ be a two-variable function for which its partial derivatives exist on the domain S . If f has a relative extremum or saddle point at $(x_0, y_0) \in S$, then

$$f_x(x_0, y_0) = 0 \text{ and } f_y(x_0, y_0) = 0.$$

Theorem 5.4.2 can be understood easily from the diagrams in Figures 5.4.1 to 5.4.3. The point $(x_0, y_0, f(x_0, y_0))$ that satisfies the equality in Theorem 5.4.2 is called **critical point**. To determine whether a critical point is a relative minimum, or relative maximum, or saddle point, we use the following theorem.

Theorem 5.4.3 (Second Partials Test)

Let $f(x, y)$ be a two-variable function for which its partial derivatives exist on the domain S . Assume that $f_{xy} = f_{yx}$ in S . Suppose f has a critical point at $(x_0, y_0) \in S$. We define the discriminant of f , denoted by D_f , as

$$D_f = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - (f_{xy})^2$$

If $D_f(x_0, y_0) < 0$, then f has a **saddle point** at (x_0, y_0) .

If $D_f(x_0, y_0) > 0$ and:

- either $f_{xx}(x_0, y_0) > 0$ or $f_{yy}(x_0, y_0) > 0$, then f has a **relative minimum point** at (x_0, y_0)
- either $f_{xx}(x_0, y_0) < 0$ or $f_{yy}(x_0, y_0) < 0$, then f has a **relative maximum point** at (x_0, y_0)

Example 5.4.4

Find the relative extrema and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 5$.

$$f_x = 4x^3 - 4y \text{ and } f_y = 4y^3 - 4x$$

$$f_x = 0 \text{ and } f_y = 0 \text{ gives a system of equations: } \begin{cases} x^3 - y = 0 \\ y^3 - x = 0 \end{cases}$$

By substitution, we have $(x^3)^3 - x = 0 \Rightarrow x(x^8 - 1) = 0 \Rightarrow x(x - 1)(x + 1)(x^2 + 1)(x^4 + 1) = 0$. So, the critical points are located at $(0, 0)$, $(1, 1)$, and $(-1, -1)$.

$$D_f = \begin{vmatrix} \frac{\partial}{\partial x}(4x^3 - 4y) & \frac{\partial}{\partial y}(4x^3 - 4y) \\ \frac{\partial}{\partial x}(4y^3 - 4x) & \frac{\partial}{\partial y}(4y^3 - 4x) \end{vmatrix} = \begin{vmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{vmatrix} = 144x^2y^2 - 16$$

Location	Discriminant	Conclusion
$(0, 0)$	$D_f(0, 0) = -16 < 0$	f has a saddle point $(0, 0, 5)$
$(1, 1)$	$D_f(1, 1) = 128 > 0$	$f_{xx}(1, 1) = 12 > 0 \Rightarrow f$ has a rel. min. point $(1, 1, 3)$
$(-1, -1)$	$D_f(-1, -1) = 128 > 0$	$f_{xx}(-1, -1) = 12 > 0 \Rightarrow f$ has a rel. min. point $(-1, -1, 3)$

In the case where f has zero discriminant at the critical point, further analysis is needed to determine whether the critical point is a relative maximum/minimum, or a saddle point.

Example 5.4.5

Find the relative extrema and saddle points of $f(x, y) = x^4 - y^4$.

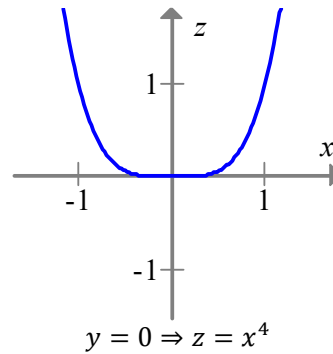
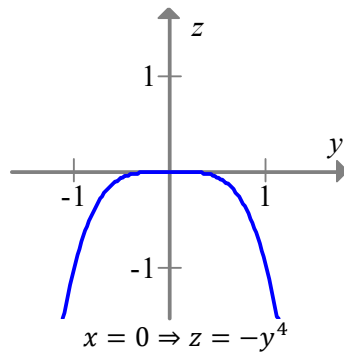
$$f_x = 4x^3 \text{ and } f_y = -4y^3$$

$f_x = 0$ and $f_y = 0$ gives a system of equations: $\begin{cases} 4x^3 = 0 \\ -4y^3 = 0 \end{cases}$, which implies $x = y = 0$.

So, f has a critical point $(0, 0, 0)$.

$$D_f = \begin{vmatrix} \frac{\partial}{\partial x}(4x^3) & \frac{\partial}{\partial y}(4x^3) \\ \frac{\partial}{\partial x}(-4y^3) & \frac{\partial}{\partial y}(-4y^3) \end{vmatrix} = \begin{vmatrix} 12x^2 & 0 \\ 0 & -12y^2 \end{vmatrix} = -144x^2y^2$$

Since $D_f(0, 0) = 0$, the second partials test does not apply. We analyze the behavior of f by observing the curves of intersection between the surface $z = f(x, y)$ and each of the planes $x = 0$ and $y = 0$ as shown below.



Thus, we conclude that f has a saddle point $(0, 0, 0)$.

5.5 Exercises

1. Calculate the partial derivatives f_x and f_y of each of the following functions.

(a) $f(x, y) = (2x + 2y)^{10}$

(e) $f(x, y) = \frac{x - y}{x + y}$

(b) $f(x, y) = \ln(x + y)$

(f) $f(x, y) = (x + y)(xy - 1)$

(c) $f(x, y) = x^y$

(g) $f(x, y) = \sin^2(x - 3y)$

(d) $f(x, y) = e^{x^2 + y^2}$

(h) $f(x, y) = e^{-x} \sin(x + y)$

2. Assuming all partial derivatives exist, use the chain rule to prove each of the following.

(a) Given $f(u, v, w)$, where $u = x - y$, $v = y - z$, and $w = z - x$, show that

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 0.$$

(b) Given $z = f(x, y)$, where $x = r \cos \theta$ and $y = r \sin \theta$, show that

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2.$$

(c) Given $z = f(x, y)$, where $x = u + v$ and $y = u - v$, show that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial u} \cdot \frac{\partial z}{\partial v}.$$

(d) If $z = f(x + ay) + g(x - ay)$, show that $\frac{\partial^2 z}{\partial y^2} = a^2 \frac{\partial^2 z}{\partial x^2}$.

3. Verify each of the following (k is a non-zero constant).

(a) $f(x, t) = e^{-t} \cos \frac{x}{k}$ satisfies the **heat equation** $\frac{\partial f}{\partial t} = k^2 \frac{\partial^2 f}{\partial x^2}$.

(b) $f(x, t) = \sin 3kt \sin 3x$ satisfies the **wave equation** $\frac{\partial^2 f}{\partial t^2} = k^2 \frac{\partial^2 f}{\partial x^2}$.

(c) $f(x, t) = \frac{e^{-x^2/(4kt)}}{\sqrt{t}}$ satisfies the **diffusion equation** $\frac{\partial f}{\partial t} = k \frac{\partial^2 f}{\partial x^2}$.

(d) The pair: $u(x, y) = e^{-x} \sin y$ and $v(x, y) = e^{-x} \cos y$ satisfies the **Cauchy-Riemann equation** $u_x = v_y$ and $u_y = -v_x$.

4. Calculate the gradient of each of the following functions at the specified point.

(a) $f(x, y) = \ln(x^2 + y^2)$ at $(1, 1)$

(b) $f(x, y, z) = x^2 + y^2 - 2z^2 + z \ln x$ at $(1, 1, 1)$

(c) $f(x, y, z) = 3e^x \cos(yz)$ at $(0, 0, 0)$

(d) $f(x, y, z) = (x^2 + y^2 + z)^{-1/2} + \ln(xyz)$ at $(-1, 2, -1)$

5. Construct the standard equation of the tangent plane and symmetric equation for the normal line to each of the following surfaces at the specified point on the surface.
- | | |
|---|---|
| (a) $z = \ln(x^2 + y^2)$ at $(1, 0, 0)$ | (d) $x^2 + y^2 + z^2 = 3$ at $(1, 1, 1)$ |
| (b) $z = e^{-(x^2 + y^2)}$ at $(0, 0, 1)$ | (e) $x^2 - 2y^2 + z^2 + yz = 2$ at $(2, 1, -1)$ |
| (c) $z = 4x^2 + y^2$ at $(1, 1, 5)$ | (f) $yz = \ln(x + z)$ at $(0, 0, 1)$ |
6. Find the relative extrema and saddle points of each of the following functions.
- | |
|---|
| (a) $f(x, y) = x^2 + y^2 + x^2y + 4$ |
| (b) $f(x, y) = e^{4y - x^2 - y^2}$ |
| (c) $f(x, y) = 2x^3 + 2y^3 - 9x^2 + 3y^2 - 12y$ |
| (d) $f(x, y) = x^4 + y^4 + 4xy$ |
| (e) $f(x, y) = x^2y^4$ |
| (f) $f(x, y) = x^2y^3$ |
7. Suppose a three-variable function $F(x, y, z)$ has continuous first and second partial derivatives. The **Laplacian operator** ∇^2 on F is defined as
- $$\nabla^2 F = \nabla \cdot \nabla F = F_{xx} + F_{yy} + F_{zz}$$
- A three-variable function F is called **harmonic** in the domain D if $\nabla^2 F = 0$ in D . Determine whether each of the following functions is harmonic on its domain.
- | |
|------------------------------------|
| (a) $F = e^{-x} \sin y$ |
| (b) $F = xyz$ |
| (c) $F = e^{-x}(\cos y - \sin y)$ |
| (d) $F = (x^2 + y^2 + z^2)^{-1/2}$ |

